



FACULTY OF COMPUTING

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SECP3123 - BUSINESS INTELLIGENCE

SECTION 2

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PROJECT PROPOSAL

TITLE:

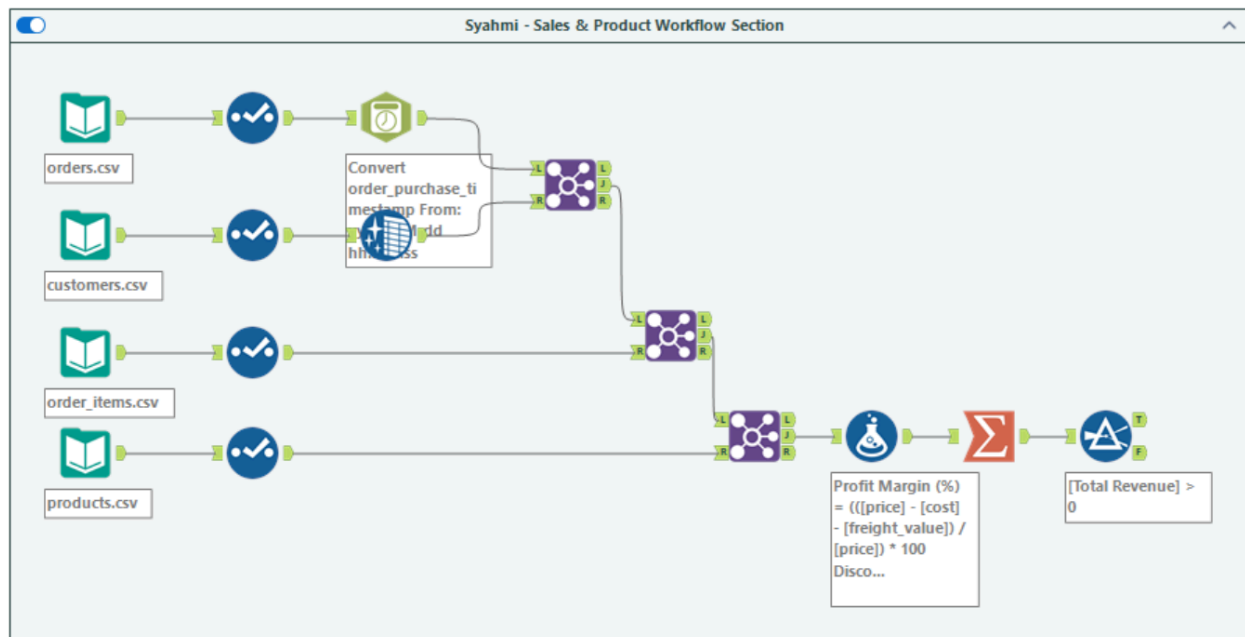
**BUSINESS INTELLIGENCE ANALYTICS FOR U.S. E-COMMERCE PERFORMANCE
OPTIMIZATION: A MULTI-DIMENSIONAL ANALYSIS OF SALES, CUSTOMER
BEHAVIOUR, AND DELIVERY OPERATIONS (2019–2025)**

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1.0 Module Sales and Product Analytics: Syahmi Faris

1.1 Screenshot of Workflow



Four CSV files: orders.csv, order_items.csv, customers.csv, and products.csv are loaded into Alteryx using Input Data tools. Only the relevant columns are kept using the Select tool, while the Data Cleansing tool removes nulls, duplicates, and whitespace issues. The order_purchase_timestamp column is also converted from plain text into a proper date format using the DateTime tool so the year can be extracted later.

The cleaned datasets are then combined into one unified table using three chained Join tools which first linking orders with customers via customer_id, then with order items via order_id, and finally with products via product_id. A Formula tool then created three new calculated columns: profit Margin (%), Discount Rate (%), and Order Year. Finally, a Summarize tool groups the data by product category, year, and customer state to calculate total revenue, and a Filter tool removes any rows where total revenue is zero.

1.2 Alteryx Tools Used

Tool	Purpose
Input Data	Load CSV files into the workflow
Select	Keep only relevant columns
DateTime	Convert timestamp text into proper date format

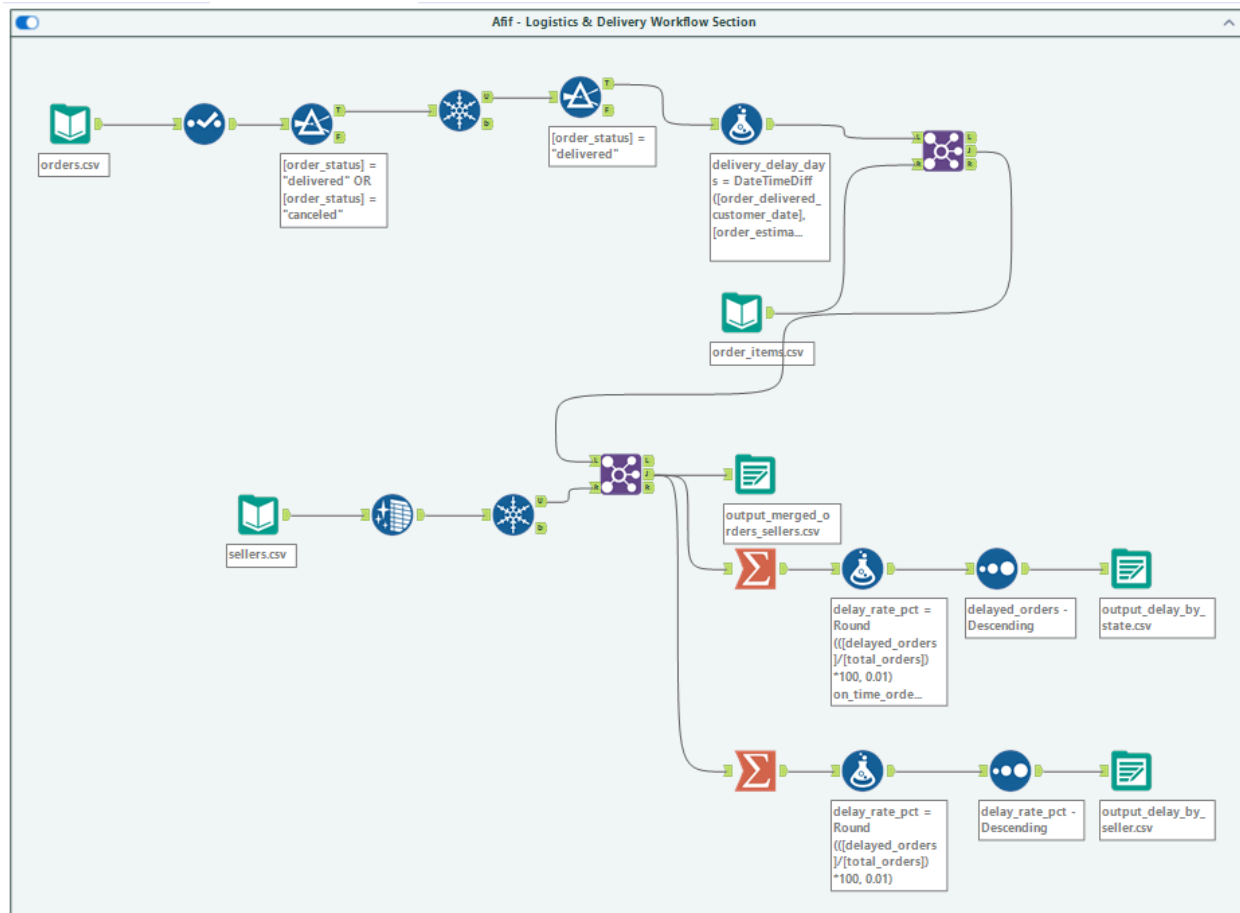
Data Cleansing	Remove nulls, duplicates, and whitespace
Join	Combine datasets using shared keys
Formula	Create Profit Margin, Discount Rate, and Order Year
Summarize	Group and aggregate total revenue
Filter	Remove zero or invalid revenue records

1.3 Challenges Faced

1. **Date format issue:** The order_purchase_timestamp was stored as text, not a date. Had to use the DateTime tool to convert it before extracting the year.
2. **Duplicate column names after Join:** After joining multiple datasets, some columns like price and order_id appeared twice with Right_ prefix. Had to carefully deselect duplicates in the Join tool's Select configuration.
3. **Data type mismatch:** Columns like price, cost, and freight_value were originally loaded as text (V_String). Had to manually change them to Double type in the Select tool before using them in the Formula tool.

2.0 Module Logistics and Delivery Analytics: Afif Shaqir Irfan

2.1 Screenshot of Workflow



Workflow Objective and Data Processing

This workflow is designed to streamline the tracking and analysis of logistics and delivery performance. It ingests raw data from customer orders, order items, and seller profiles, transforming scattered files into a reliable, consolidated dataset. The initial processing phase focuses on structural cleaning: standardizing date formats, filtering out canceled or invalid transactions, and handling missing seller information. A core function of this stage is calculating delivery variances specifically measuring the exact number of days a package arrives early or late compared to its estimated delivery date.

Performance Aggregation and Insights

Once the data is unified, the workflow aggregates the information to evaluate fulfillment efficiency. It groups the consolidated data by geographic state and individual seller, automatically calculating key performance metrics such as total order volume, average delivery times, and specific delay rate percentages. This automated analysis eliminates manual

calculations and quickly highlights operational bottlenecks. The final outputs provide a clear, data-driven overview of the supply chain, making it easy to identify high-performing regions and pinpoint specific sellers that require operational improvements.

2.2 Alteryx Tools Used

Tool	Purpose
Input Data	Load the orders.csv , sellers.csv and order_items.csv files into the workflow.
Select	Convert text data types into proper DateTime formats for the date columns. (example: purchase, delivery and estimated dates).
Filter	Filter the data to keep only valid orders (specifically those with a “delivered” or “canceled status”).
Data Cleansing	Clean the seller information by removing leading whitespaces and fixing null or empty values.
Formula	Create new columns to calculate the delivery delay days (delivery_delay_days), a delayed flag (is_delayed as 1 or 0), and the delay rate percentage (delay_rate_pct)
Join	Combine the Orders, Order Items, and Seller datasets into a single unified table using shared keys (order_id and seller_id)
Sort	Order the summary tables in descending order to quickly identify States or Sellers with the highest number or percentage of delays.
Output Data	Export and save the final merged datasets and the summarized tables into new CSV files.

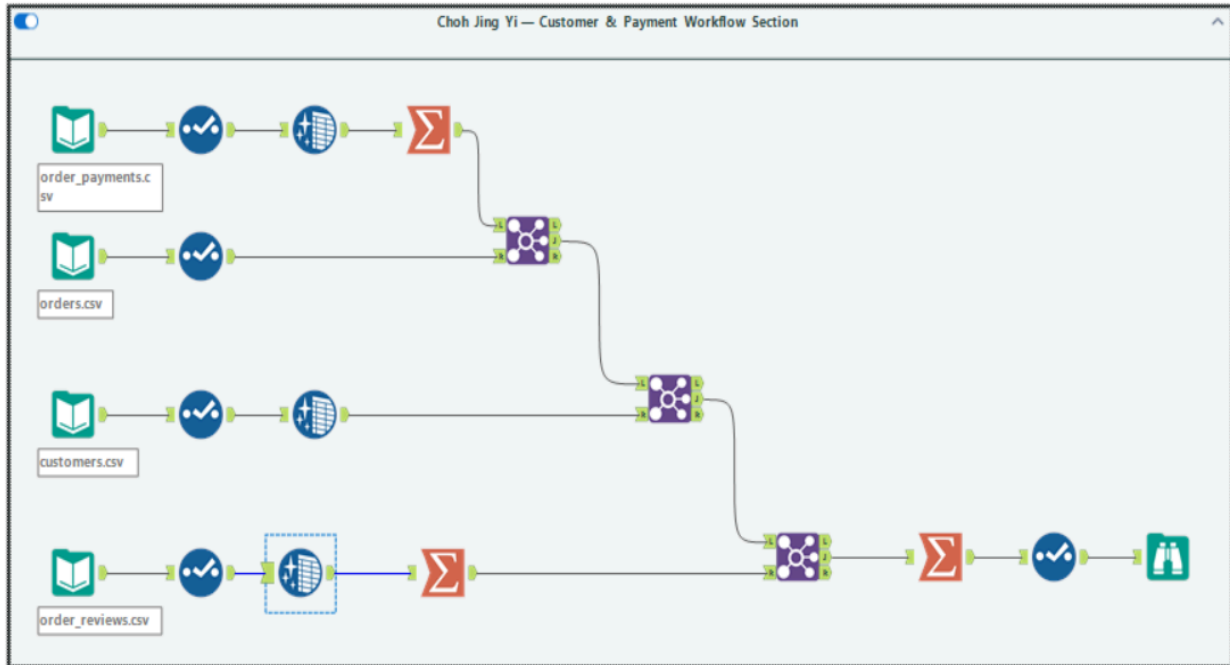
2.3 Challenges Faced

1. **Linking the Data:** At first, the main orders file did not include the seller IDs. An extra file (the order items list) needs to be added to act as a bridge. This allowed us to successfully connect each customer order to the correct seller.
2. **Counting Late Deliveries:** The system initially struggled to count only the delayed orders, accidentally counting all of them instead. This is fixed by changing the "late" label into a number (1 for late, 0 for on-time) so the software could add them up correctly.
3. **Understanding Negative Numbers:** When calculating the average delivery time, the final results were showing up as negative numbers. We had to analyze the data to

understand that a negative number was actually a good thing because it meant the packages were arriving earlier than the estimated date.

3.0 Module Customer and Payment Analytics: Choh Jing Yi

3.1 Screenshot of Workflow



First, four datasets (customers.csv, orders.csv, order_payments.csv, and order_reviews.csv) were loaded. Because customers and payments do not share a direct ID, the orders.csv table was used as a bridge to connect them.

Before joining the tables, it uses the Summarize tool on the payments and reviews data. Since one order can have multiple payments like split installments, joining them directly would create duplicate rows. By grouping the payments by order_id first, the data remained accurate. Finally, the tables were merged using Join tools, and a final Summarize tool was used to find the average review score and total revenue for each payment method.

3.2 Alteryx Tools Used

Tool	Purpose
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Input Data	Load CSV files into the workflow
Select	To keep only the necessary columns and change text data into numbers
Data Cleansing	To remove empty rows, delete duplicates, and clean up extra spaces in the text.
Summarize	Applied summarize 3 times to ensure accurate data grouping and calculations: 1. Grouped payments by order_id and summed the payment values to prevent duplicate rows before joining. 2. Grouped reviews by order_id and averaged the review scores. 3. Grouped the final merged dataset by payment_type to calculate the total orders, average customer satisfaction, and total revenue.
Join	To connect the different datasets together using shared keys (order_id and customer_id)
Browse	To view the final, cleaned results before exporting

3.3 Challenges Faced

1. **Data Duplication via One-to-Many Relationships:** The raw order_payments.csv dataset contained multiple rows for a single order if a customer uses split payments or multiple installments. Joining this directly to the main orders table would have resulted in severe data duplication and skewed metrics.
2. **Data Type Mismatches:** Some fields such as payment_value and review_score were initially imported as plain text strings (V_String). Because they were not recognized as numbers, it was impossible to perform required mathematical aggregations like Sum and Average.